

KAI-HSIN HUNG

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EDUCATION

UNIVERSITY OF THE PACIFIC, Stockton, California

May 2026

Master of Science in Data Science, GPA: 3.96 / 4.00

Relevant Coursework: Relational Databases, Machine Learning, Natural Language Processing

MING CHI UNIVERSITY OF TECHNOLOGY, New Taipei City, Taiwan

Jun. 2023

Bachelor of Science in Materials Science and Engineering

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Databases & Data: PostgreSQL, PostGIS, SQLAlchemy, Alembic, Pandas, NumPy, JSON/CSV Schema Design

Cloud & DevOps: Docker, Docker Compose, Google Cloud (Cloud Run, Firebase), GitHub Actions (CI/CD), REST APIs

Frameworks & Tools: FastAPI, Flask, Scikit-Learn, PyMuPDF, BeautifulSoup, Git

EXPERIENCE

BirthdayMessaging.io, London, UK (Remote)

Mar. 2025 – May 2025

Software Engineer Intern

- Engineered a mass data ingestion pipeline and a custom .VCF file parser in Python to extract, transform, and load 1,000+ bulk contact records into the application database
- Architected an idempotent, cron-based API task scheduler that queries time-sensitive database events and automates email dispatch, eliminating duplicate message deliveries
- Implemented a transactional logging system that captures API response payloads and writes structured execution records to the database, enabling real-time observability of background job execution

PUBLICATION

K. Hung, G. Phung, F. Lorenzino, "GoPark: An AI-Powered Parking Recommendation System," *Intelligent Systems Conference (IntelliSys) 2026*, Springer Lecture Notes in Networks and Systems. (Accepted)

SELECTED PROJECTS

Toxicity Prediction Data Pipeline (Capstone)

May 2026

Python, Gemini API, REST APIs, Pandas, Scikit-Learn, JSON/CSV

- Architected an automated ETL pipeline to ingest, clean, and standardize 10 heterogeneous healthcare datasets (LIVERTOX, SIDER, DrugBank, Clinical Trials) from structured and unstructured sources into a unified JSON/CSV schema
- Harmonized TG-GATEs and DrugMatrix datasets (~22 GB combined) containing gene expression, clinical chemistry, and pathology data, standardizing dose units, duration formats, and outcome labels to produce model-ready outputs
- Implemented automated data quality checks, lineage tracking, and validation rules across all curated datasets, ensuring identifier traceability (CAS, InChIKey, DrugBank ID) and cross-source consistency
- Developed a scalable text-processing workflow utilizing the Gemini API to parse unstructured medical abstracts, outputting structured JSON validation records with automated quality checks

GoPark: AI-Powered Parking Recommendation System

Dec. 2025

Python, FastAPI, PostgreSQL, PostGIS, SQLAlchemy, Alembic, Docker, SwiftUI

- Designed and implemented a PostgreSQL/PostGIS database schema with SQLAlchemy ORM and Alembic migrations, modeling spatial stall geometries, occupancy states, and vehicle event logs with referential integrity constraints
- Built a containerized FastAPI backend with Docker Compose, orchestrating API, database, and ML model services into a reproducible multi-container deployment
- Engineered an end-to-end pipeline from raw image ingestion through object detection to recommendation ranking, achieving sub-50 ms average latency on edge hardware

DeepTSMC: Automated Time-Series Forecasting Pipeline

Jun. 2025

Python, TensorFlow, Pandas, Scikit-Learn, Flask, React, Docker, CI/CD

- Developed an automated data pipeline using Pandas and Scikit-Learn for imputation, scaling, and sequence windowing to supply normalized time-series inputs to deep learning models
- Built a model-serving infrastructure using Docker, Google Cloud Run, and GitHub Actions CI/CD, reducing deployment time by 85% and enabling continuous delivery
- Designed a containerized frontend-backend architecture with Flask REST API and React dashboard, orchestrated via Docker Compose with automated health checks and environment-based configuration.